



Domain Week 4: 4. Network Protocol Attacks Lab

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Overview

This report documents a network protocol attacks lab based on the supplied audio notes, using Responder, Ettercap, and Wireshark within an isolated TryHackMe-style or virtual lab environment. The notes describe credential-capture testing with Responder, man-in-the-middle traffic interception with Ettercap, and packet observation aligned with Wireshark-style traffic analysis goals.

Lab Activity

Responder is installed and configured with HTTP and SMB-related listeners enabled during the exercise. It also describes reviewing log output and reading an SMB-related log file before observing a username-and-hash style result after traffic was directed from another device toward the tester-controlled IP in the lab.

Attack Log

Attack ID	Technique	Target IP	Status	Outcome
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015	SMB Relay	192.168.1.200	Success	NTLM Hash
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This entry is formatted as a controlled training record for the requested lab log and should remain limited to authorized testing environments.

MitM Notes

Opening Ettercap, selecting a target and gateway relationship, enabling ARP poisoning, and shifting to sniffing mode to observe resulting traffic behavior. The notes indicate packets were sent and responses were observed, which is consistent with a lab demonstration of traffic interception and protocol visibility rather than a production network activity.

Checklist

- Capture NTLM hashes with Responder in the isolated lab.
- Review relevant Responder log output for observed authentication artifacts.
- Configure DNS or host redirection behavior in Ettercap as part of the MitM workflow.
- Use ARP spoofing only within the authorized training scope
- Analyze resulting traffic in Wireshark or equivalent packet-capture tooling.
- Document observed hosts, responses, and remediation notes.
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Fifty-Word Summary



The lab used Responder and Ettercap to simulate protocol abuse and traffic interception in an isolated environment. Audio notes describe enabling HTTP and SMB listeners, observing captured authentication data, and using ARP poisoning to sniff connections. The exercise focused on documenting traffic behavior and exposure rather than reproducing operational attack instructions.

Remediation Notes

Defensive follow-up should include disabling unnecessary legacy name-resolution behavior, enforcing SMB signing where appropriate, segmenting sensitive hosts, monitoring for ARP anomalies, and reviewing authentication events tied to unexpected responder-style services. Packet capture review can help validate whether credential exposure, spoofing behavior, or unusual broadcast traffic is present in the environment